SECTION 1 · SUBSECTION 1.1
BHATKHANDE-FLOW RENDERING ENGINE

1.1.1 INTRODUCTION.

Bhatkhande-Flow is a rendering engine for Bhatkhande notation. It reads a bandish as JSON, swartaal XML or a Label Studio export and draws the printed page as SVG. Every drawn element carries a BhatkhandeSet class name, so one run also yields training labels (figure 1.1-1). The pipeline is VexFlow's: measure first, place second, draw through an abstract context.

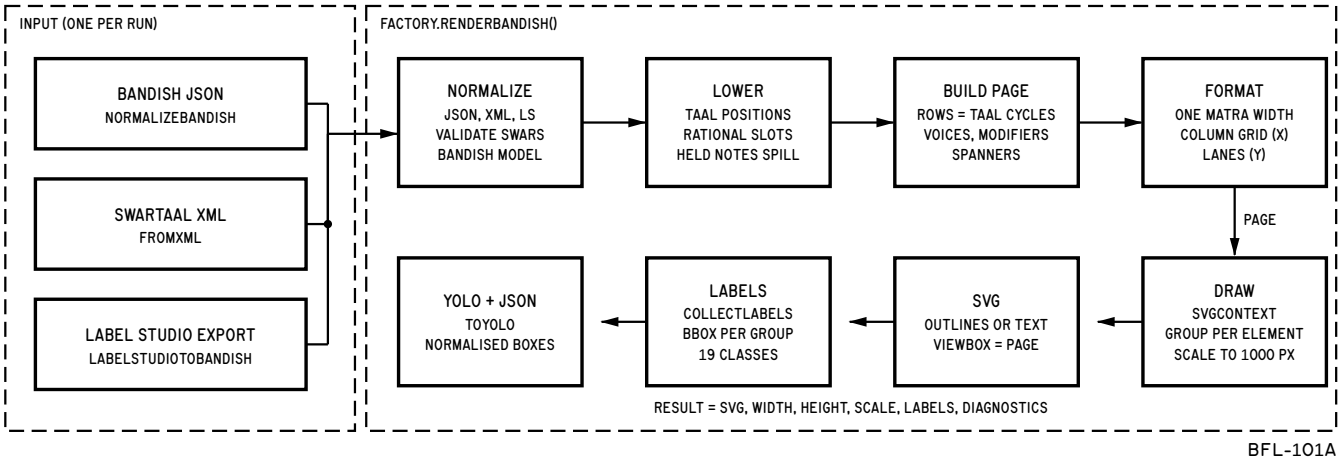


Figure 1.1-1. Rendering Pipeline Functional Diagram

1.1.2 FUNCTIONAL DESCRIPTION.

Factory.renderBandish() sets the style and fonts, parses the input, lowers it to rational slots, builds one row per taal cycle and formats the whole page. Page.draw() then draws the header, the rows and the spanners into an SVGContext, which records a bounding box for every element group. Source size by module is shown in figure 1.1-2.

1.1.2.1 One Page, One Width.

Every row is measured first. The page then takes one matra width: the widest packed matra, or 1.25 times the swar size if larger, plus 7 px of padding each side. A page wider than 1000 px is scaled down as a whole. Rows never wrap.

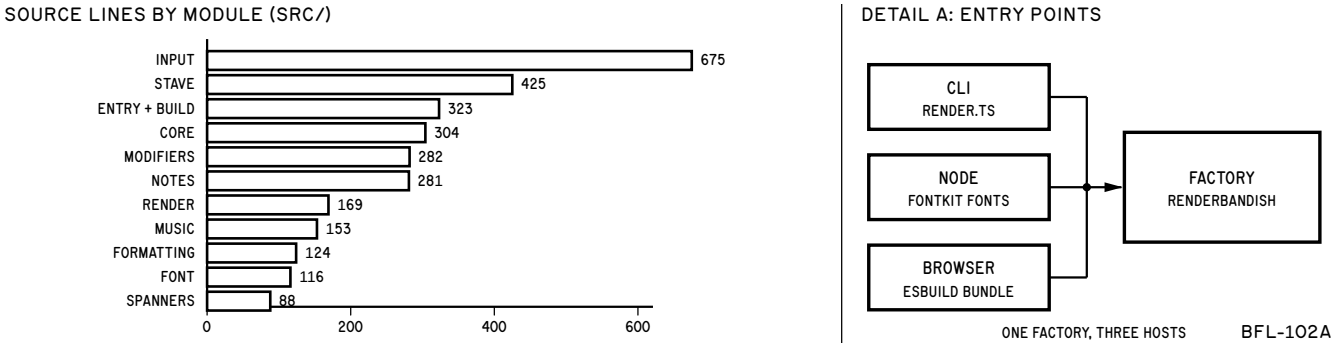


Figure 1.1-2. Source Modules And Entry Points

1.1.3 PERFORMANCE AND DESIGN DATA.

Table 1.1-1. Engine Data

Parameter	Value
Source files (TypeScript)	53 FILES, 2,940 LINES
Test suite	18 TESTS, ALL PASS
Taal definitions	11
Label classes	19
Example bandish	7 JSON, 7 LABEL STUDIO
Page width limit	1000 PX, THEN SCALED
Swar / lyric font size	24 / 21 PX
Output per render	SVG, LABELS, YOLO (+ PNG)

CAUTION

Run *npm install* on the machine that runs the code. A node_modules folder built on Linux carries the wrong esbuild binary,.

RENDER A BANDISH — USER

```
1 SHELL - cd bhatkhande-flow
2 SHELL - npm install
3 SHELL - npm test
  √ tests - 18 pass
4 SHELL - npm run render --
    examples/bhimpalasi-geet.json
    out --png
  √ out - svg, labels, yolo, png
```

BFL